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# Enhanced UPI Fraud Detection Using CNN: A Comparative Analysis with Machine Learning Models

**Sekuri Manju Bhargavi [1*], Bandaru Kesava Ram [2]**
[1-2] Student, Department of Information Technology, Kallam Haranadhareddy Institute of Technology, Chowdavaram, Guntur, Andhra Pradesh, India-522019

* Corresponding Author: **Sekuri Manju Bhargavi**

## Article Info

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**Abstract**
The Unified Payment Interface (UPI) has transformed the landscape of digital payments, significantly enhancing the pace of digitization. It has redefined the financial ecosystem by facilitating quicker, simpler, and more accessible cashless transactions for individuals across the nation. However, the increasing use of UPI has also made it a target for various fraudulent activities. Therefore, it is essential to detect fraudulent transactions in real-time by leveraging artificial intelligence techniques. Many machine learning algorithms are used to detect fraudulent transactions, but most of them rely heavily on historical transactional data to identify patterns and anomalies. While historical data provides valuable insights into past fraud patterns, it may not be sufficient to detect dynamic patterns in evolving fraudulent transactions. We propose a CNN-based fraud detection model that recognizes complex patterns in real-time transactional data. In this paper, we analyze the performance of machine learning algorithms and a CNN-based approach. Our results demonstrate that the CNN-based model outperforms traditional machine learning models in detecting UPI frauds.

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**Keywords:** UPI, Fraud Detection, Machine Learning, Convolution Neural Networks, Fraudulent Transactions

## Introduction

Unified Payments Interface (UPI) has emerged as one of the prominent and widely adopted digital payment systems [1, 3]. Its simplicity, interoperability, and ease of use have contributed to its rapid growth, making it a preferred mode of payment for millions of users [4]. However, the increasing volume of UPI transactions has also made it an attractive target for various fraudulent activities, including phishing, social engineering, and unauthorized access. Traditional fraud detection models face significant challenges due to the dynamic nature of these frauds, which are characterized by changing tactics and patterns [5].

Traditional machine learning models, such as logistic regression, decision trees, random forests, and support vector machines (SVM), have been widely used for fraud detection [6]. These models typically rely on historical transactional data to identify patterns and anomalies. They are effective to some extent, but often struggle to detect evolving fraud tactics due to their dependency on past data and static pattern recognition [7]. Moreover, the high dimensionality and complexity of real-time transactional data require more advanced techniques to capture dynamic behaviors.

To address these challenges, deep learning models, such as CNNs are used because of their ability to recognize complex patterns in high-dimensional data. CNNs, have demonstrated significant potential in financial applications due to their capability to analyze spatial and temporal dependencies in data. In this paper, we propose a CNN-based fraud detection model that uses real-time transactional data to identify complex fraud patterns with high accuracy. The proposed model is compared with traditional machine learning approaches, and the results demonstrate that the CNN-based model outperforms conventional methods in detecting UPI frauds.

## Related Work

This section provides a review of existing approaches, highlighting their advantages and limitations.

Rupa Rani *et al* [8] proposed XGBoost algorithm for fraud detection. It gives accuracy of 98.2%. The main limitation of this model is that it relies on static datasets and fixed patterns for fraud detection, which may reduce its effectiveness in identifying evolving fraud strategies.

Bohdan Mytnyk *et al* [9] explained various machine learning algorithms to improve fraud detection accuracy. This work demonstrates various machine learning models used for detecting fraudulent transactions, highlighting their advantages and limitations. This work specifically highlights improved results with logistic regression, decision tree, and random forest algorithms. The major limitation of this work is that it does not consider dynamic patters from real-time transactional data.

S. K. Lokesh Naik *et al* [10] recommended leveraging artificial intelligence and machine learning to effectively detect fraudulent UPI transactions. They proposed techniques for fraud detection using public data, managing imbalanced data, adapting to evolving fraud patterns, and reducing false alarms. The practical implementation of AI and machine learning algorithms not accurately justified.

## Proposed Method

In this work, we propose a CNN based model to effectively detect fraudulent transactions and send alert message to the user. The proposed model is trained with UPI transactional dataset containing various attributes like payer, payee, date, category, Reference No., Type of Transaction, Amount, Location etc the proposed methodology is illustrated in figure 1. This process consists of the following steps.

a) **User Authentication:** The system verifies the identity of the user.
b) **Transaction Processing:** UPI transactions are processed. This stage includes data collection and initial validation.
c) **Feature Engineering:** The system extracts important features from the transaction data.
Examples of features: transaction amount, frequency, location, device used, etc. Proper feature extraction improves the accuracy of fraud detection models.
d) **Fraud Detection Models:** The extracted features are passed to CNN model for fraud detection.
e) **Fraud Scoring:** The models produce a fraud score
f) **Real-time decision making:** Based on the fraud score, the system decides whether the transaction is legitimate or fraudulent.
g) **Security Actions:** If a transaction is flagged as fraudulent, the system sends an alert message to user.

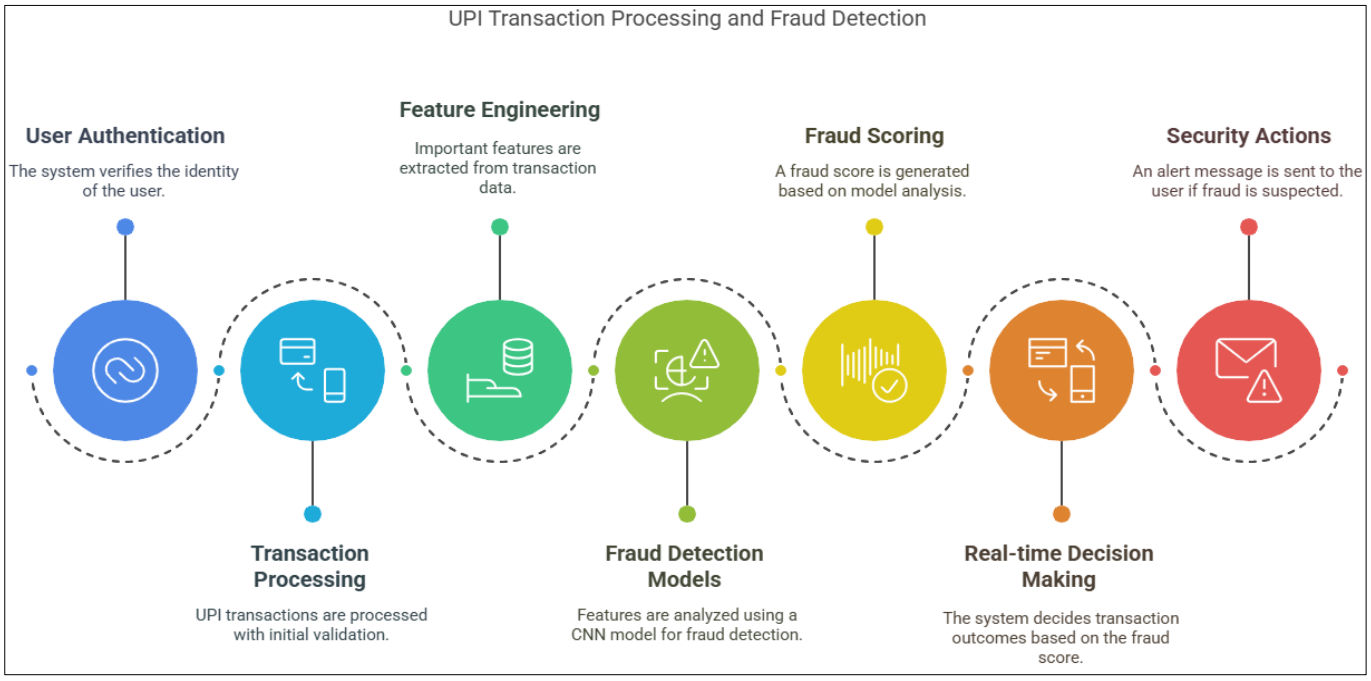



**Fig 1:** Architecture of CNN-based model for Fraud Detection

In this work, we applied CNN model at Fraud Detection Phase that extracts the features and hidden patterns from the transactional dataset to confirm a transaction as legitimate or fraudulent. The proposed model is designed by applying following steps.

**Step 1:** Input Data: Transaction data represented in matrix format.

**Step 2:** Convolution Layers: Extracts required patterns and features from the dataset. Conv1D layers used to capture temporal and feature based patterns.

**Step 3:** Pooling Layer: Maxpooling1D applied to reduce dimensionality and extract required features.

**Step 4:** Flatten Layer: converts multidimensional feature maps into 1D vector.

**Step 5:** Fully Connected Layer: Combines all the extracted features to generate a decision.

**Step 6:** Sigmoid Function: It is used to generate Fraud Score.

The CNN model for fraud detection is depicted in Figure 2.

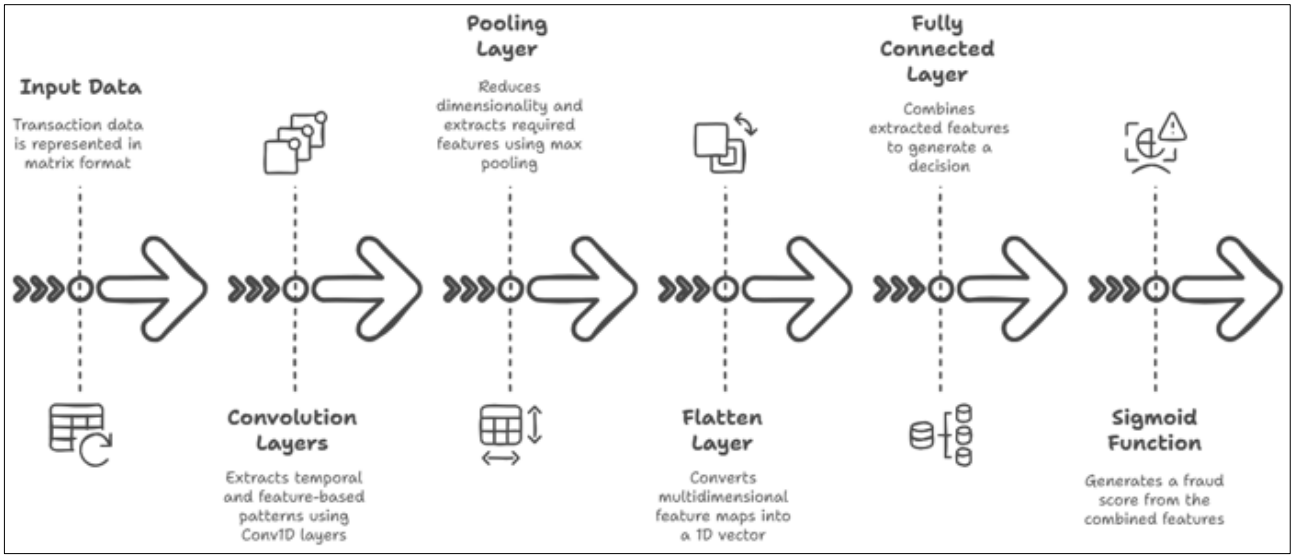



**Fig 2:** Fraud Detection Process using CNN model

## Results and Discussion

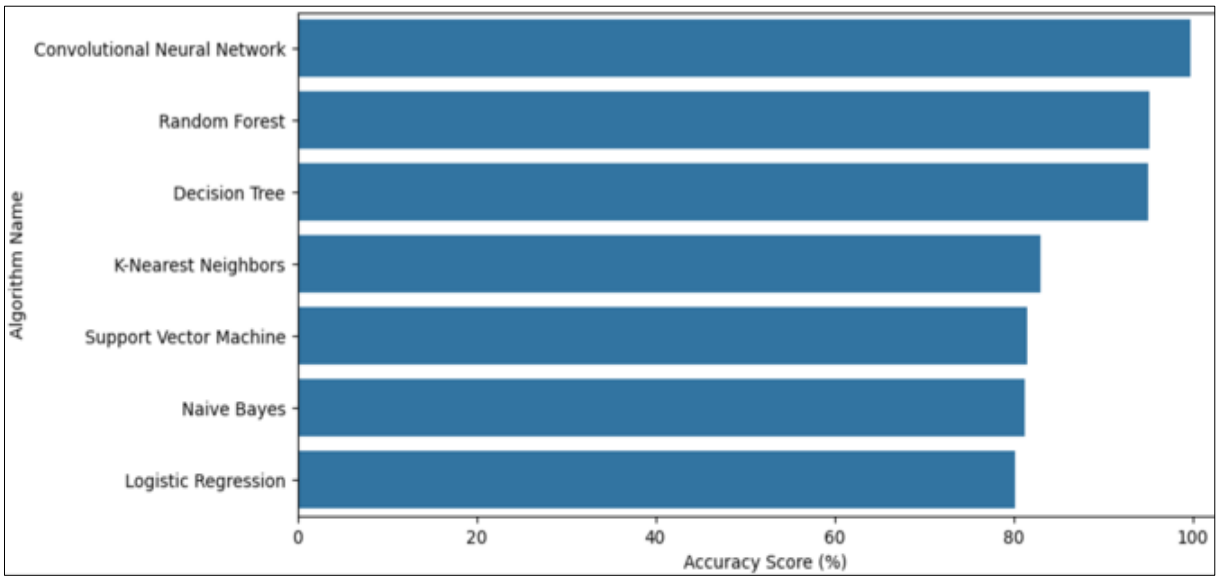
We used UPI transaction dataset from kaggle to train our proposed model. We split the dataset as 70% for training, 10% for validation and 20% for testing. The following metrics are considered to evaluate the performance of our CNN-based using TensorFlow.

- Accuracy
- Precision
- Recall
- F1-Score

Furthermore, we compared the proposed model with existing machine learning models, including Random Forest, Decision Tree, K-Nearest Neighbors, SVM, Naïve Bayes, and Logistic Regression, to analyze its performance. Table 1 presents the performance metrics of all the models considered in this study. Figure 3 illustrates the comparative analysis of existing and proposed models.

**Table 1:** Comparative Analysis of Machine Learning and Proposed CNN Model

| Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|
| Logistic Regression | 0.802 | 0.9081 | 0.73 | 0.80 |
| K-Nearest Neighbor | 0.834 | 0.834 | 0.878 | 0.855 |
| Support Vector Machine | 0.815 | 0.958 | 0.708 | 0.815 |
| Decision Tree | 0.95 | 0.937 | 0.978 | 0.945 |
| Naïve Bayes | 0.812 | 0.969 | 0.695 | 0.810 |
| Random Forest | 0.96 | 0.969 | 0.960 | 0.965 |
| Proposed CNN Model | 0.997 | 0.856 | 0.904 | 0.879 |



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| | Algorithm Name | Accuracy Score (%) |
|---|---|---|
| 6 | Convolutional Neural Network | 99.735218 |
| 5 | Random Forest | 95.250000 |
| 4 | Decision Tree | 95.000000 |
| 1 | K-Nearest Neighbors | 83.000000 |
| 2 | Support Vector Machine | 81.500000 |
| 3 | Naive Bayes | 81.250000 |
| 0 | Logistic Regression | 80.250000 |

**Fig 3:** Comparison of accuracy scores between the proposed model and existing machine learning models.

Our results justify that the proposed model performs better than the existing machine learning-based models.

## Conclusion

We proposed a CNN-model with sigmoid to effectively detect UPI fraudulent transactions in real-time. A comparative study was also conducted to evaluate the performance of proposed model and existing machine learning models used for UPI fraud detection. Our results justify that the proposed model delivers superior performance compared to the existing models. In future, we extend this work by incorporating additional features to enhance the model's performance and better detection of fraudulent transactions with dynamic patterns.

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